Reinforcement Learning

Monte Carlo Methods

Sutton/Barto* Chapter 5

Michael Hahsler, SMU

With figures from Sutton/Barto*

*Sutton and Barto, Reinforcement Learning: An Introduction, 2nd edition, MIT Press, Cambridge, MA, 2018

SMU Lyle School of Engineering

Online Material

Topics of this Course

- Introduction to reinforcement learning
- Markov decision processes
- **Part I: Tabular Methods**
  - Dynamic programming
  - **Monte Carlo methods**
  - Temporal-difference learning
  - Multi-step bootstrapping
  - Planning and learning with tabular methods
- Part II: Approximate Solution Methods
  - Prediction and Control using Approximation
  - Eligibility Traces
  - Policy Gradient Methods
- Part III: Modern RL Methods
  - Deep Reinforcement Learning
  - Current Applications

Summary of Notation

General

X	capital letters: random variables
x, p	lower-case letters: realizations of random variables or scalar functions
$\boldsymbol{w}$	Bold lower-case letters: real-valued vectors (even if random variables)
$\boldsymbol{W}$	bold capitals: matrices
α	Greek letters: parameters (vectors if in bolt)
$\Pr\{X = x\}$	probability that a random variable X takes on the value x
$X \sim p$	random variable X selected from distribution $p(x) = \Pr\{X = x\}$
$\mathbb{E}[X]$	expectation of a random variable X, i.e., $\mathbb{E}[X] = \sum_x p(x)x$
$\text{argmax}_\text{a}\ f(a)$	a value of action a at which $f(a)$ takes its maximal value

Value Function

G_t	return (cumulative reward) following time t
$G_{t:h}$	return from t to h (discounted and corrected)
$v_\pi(s)$	value of state s under policy π (expected return)
$v_*(s)$	value of state s under the optimal policy
$q_\pi(s,a)$	value of taking action a in state s under policy π
$q_*(s,a)$	value of taking action a in state s under the optimal policy
V, V_t	array estimates of state-value function v_π or v_*
Q, Q_t	array estimates of action-value function q_π or v_*

MDP

s, s'	states
a	an action
r	a reward
$\mathcal{S}$	set of all (nonterminal) states, $\mathcal{S}^+$ are all states
$\mathcal{A}(s)$	set of all actions available in state s
γ	discount-rate parameter
t	discrete time step
T	final time step of an episode (a.k.a. horizon)
A_t	random variable for the action at time t
S_t	random variable for the state at time t
R_t	random variable for the reward at time t
$p(s', r \mid s, a)$	probability of transition to state s' and receiving reward r, from state s taking action a.
$p(s' \mid s, a)$	probability of transition to state s' fom state s taking action a.
$r(s,a)$	expected immediate reward from state s after action a.
$r(s,a,s')$	expected immediate reward from state s to s' with action a.
$\pi(a \mid s)$	probability of taking action a in state s under stochastic policy π
$\pi(s)$	action taken in state s under deterministic policy π

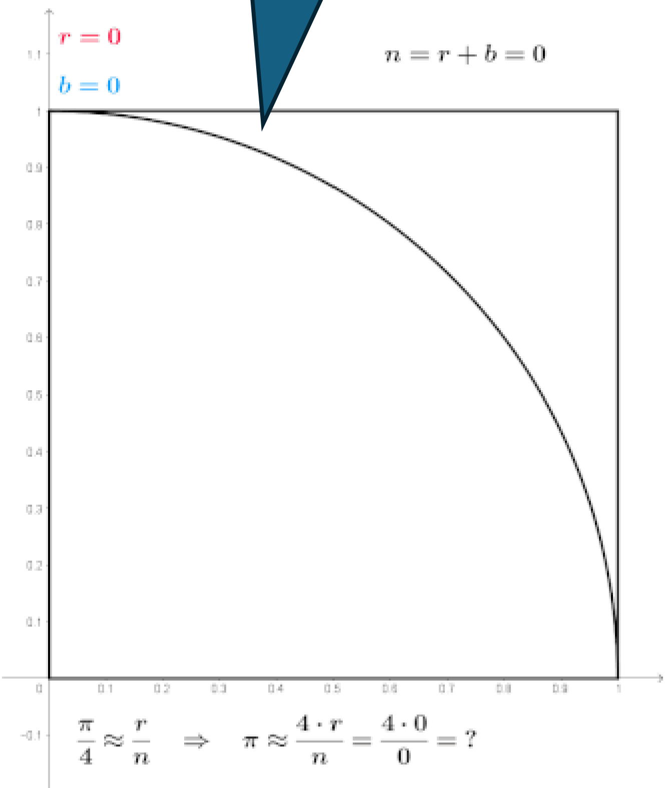
Introduction

- **Model-free method**:
  - We do not assume knowledge of the environment model (the MDP).
  - We use "experience" = **sample sequences** of states, actions, and rewards from actual or simulated interactions with the environment.
- **Remember from AI**: model-free means the agent faces an unknown, fully observable, possibly stochastic environment.
- **Advantages**:
  - We often have **sample access** to systems.
  - It is easier to simulate transitions than to specify the complete MDP.

- **Monte Carlo (MC) methods** are a family of algorithms that rely on repeated random sampling to obtain numerical results (see example to the right).

- **Idea**: Use MC to sample sequences from the environment and use them to estimate the value function.

The number of randomly sampled points is proportional to the area.



Monte Carlo method applied to approximating the value of π
Source: Wikipedia

Monte Carlo Prediction

Estimate state values by sampling with a given policy.

MC Value Function Approximation

- Use the sample-average method to approximate the expectation in:

$$v_\pi(s) = \mathbb{E}_\pi[G_t \mid S_t = s]$$

- **Convergence**: The average return of many sampled sequences (called episodes or trajectories) starting at s and following π converges to the true expectation as the number of samples increases.

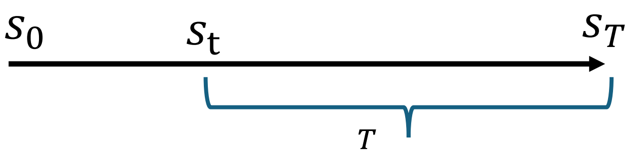
$$v_\pi(s) = \lim_{n\to\infty} \frac{1}{n} \sum_{k=1}^{n} G_{t|S_t=s,\pi}$$

- **Properties**:
  - **High Variance**: Sampling a single episode return has high variance since it contains the randomness over the whole sampled episode. Averaging over many sampled returns reduces variance.
  - **Unbiased**: The sample-average method provides an unbiased estimate (is on average correct) and thus converges for large n.
  - **Not a true online method**: This methods needs to sample whole episodes before any calculation can be done. It is not a true step-by-step online method!

MC Prediction: Estimate v_π for a given π

- **Idea:** We can update all state values along the episode.
- **First-visit**: Using the return only for the first visit of a state in the episode ensures that the sampled returns are i.i.d. distributed (not dependent on each other) and we get an **unbiased** estimate.

Episode



$$G_t = \sum_{k=0}^{T} \gamma^k R_{t+k+1}$$

Calculate the return following s_t for all states in the episode

First-visit MC prediction, for estimating $V \approx v_\pi$

Input: a policy π to be evaluated

Initialize:

  $V(s) \in \mathbb{R}$, arbitrarily, for all $s \in \mathcal{S}$

  $Returns(s) \leftarrow$ an empty list, for all $s \in \mathcal{S}$

Loop forever (for each episode):

  Generate an episode following π: $S_0, A_0, R_1, S_1, A_1, R_2, \ldots, S_{T-1}, A_{T-1}, R_T$

  $G \leftarrow 0$

  Loop for each step of episode, $t = T-1, T-2, \ldots, 0$:

    $G \leftarrow \gamma G + R_{t+1}$

    Unless S_t appears in $S_0, S_1, \ldots, S_{t-1}$:

      Append G to $Returns(S_t)$

      $V(S_t) \leftarrow \text{average}(Returns(S_t))$

We observe a sequence of rewards

Calculate remaining returns in reverse order!

Uses only the first visit.

This sample average approximates the expected return.
Note: This can be implemented with incremental updating of the averages!

Monte Carlo Control

Find a good policy while using it to sample the data.

MC Control as Generalized Policy Iteration (GPI)

- Approx. π_* using ideas from GPI with a Q-function estimated using MC prediction.

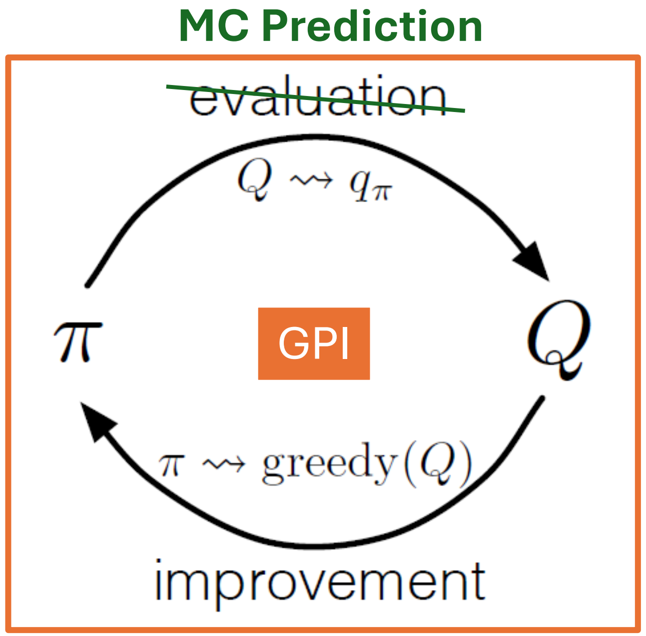
- MC policy iteration:

$$\pi_0 \xrightarrow{E} q_{\pi_o} \xrightarrow{I} \pi_1 \xrightarrow{E} q_{\pi_1} \xrightarrow{I} \pi_2 \xrightarrow{E} q_{\pi_2} \xrightarrow{I} \pi_3 \xrightarrow{E} q_{\pi_3} \xrightarrow{I} \dots \pi_* \xrightarrow{E} q_*$$

- Process:
  1. Start with an arbitrary policy.
  2. **Evaluation**: Update MC estimate by sampling some episodes using the current π.
  3. **Improvement**: $\mathrm{greedy}(Q)$ means

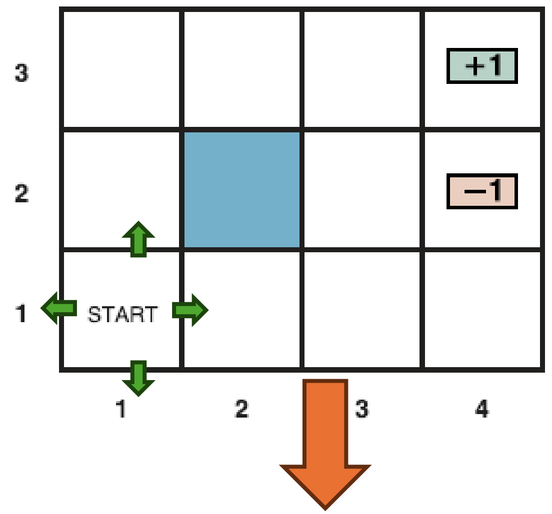
$$\pi(s) = \operatorname*{argmax}_a q(s,a) \quad \forall s \in \mathcal{S}$$

  4. Repeat evaluation and improvement till the policy does not change (= optimal policy is reached).



MC Prediction to Estimate Q-Values

- MC prediction for $q_\pi(s,a) = \mathbb{E}_\pi[G_t \mid S_t = s, A_t = a]$.
- Same as for $v_\pi(s)$, just record and average returns for action-state pairs (a,s) when following π.
- **Convergence**: Unbiased samples lead to convergence to q_π after ∞ episodes.
- **Exploration issue**: Many (a,s)-pairs that are important for π^* may never be visited when following π. To get Q-Value estimates, we need to keep exploring!
- Options to guarantee that all (a,s)-pairs are sampled :
  - **a)** **Exploring starts**: choose the starting (a,s) for each episode randomly.
  - b) Only consider **soft policies** like ϵ-greedy.
    Remember: Soft policies have non-zero probabilities for all actions so they will explore.



Q-Table

s	a	$q(s,a)$
(1,1)	up	Sample average
(1,1)	right	Sample average
(1,1)	down	Sample average
(1,1)	left	Sample average
…	…	…

Remember: Policy Improvement Theorem

If we update the MC $q(s,\cdot)$ estimate by adding more unbiased sample returns following π_k, then we can construct a new greedy policy π_{k+1} that is better (or equal) compared to π_k. Reason:

$$
\begin{aligned}
v_{\pi_{k+1}}(s) = q_{\pi_k}\big(s, \pi_{k+1}(s)\big) &= q_{\pi_k}(s, \operatorname*{argmax}_a q_{\pi_k}(s,a)) \\
&= \max_a q_{\pi_k}(s,a) \\
&\geq q_{\pi_k}\big(s, \pi_k(s)\big) \\
&\geq v_{\pi_k}(s)
\end{aligned}
$$

π_{k+1} chooses greedy action for s

Choosing the best action will never lead to a lower value!

$\rightarrow$ converges to π_*

Assumptions

- All state/action pairs are tried (e.g., start episodes with exploring states)
- Perfect policy evaluation with ∞ many episodes before each policy update. **Perfect policy evaluation is impractical!**

The effect of imperfect policy evaluation

- Individual q estimates will be wrong but unbiased return estimates will produce in expectation (on average) the correct estimate.
- Policy improvement sometimes may pick the wrong action, but **on average favors actions with true higher values.**

Alg. MC Control with Exploring Starts

So all state-action pairs are tired.

First-visit MC prediction, for estimating $V \approx v_\pi$

Input: a policy π to be evaluated
Initialize:
    $V(s) \in \mathbb{R}$, arbitrarily, for all $s \in \mathcal{S}$
    $Returns(s) \leftarrow$ an empty list, for all

Loop forever (for each episode):
    Generate an episode following π: S
    $G \leftarrow 0$
    Loop for each step of episode
        $G \leftarrow \gamma G + R_{t+1}$
        Unless S_t appears in $S_0, S_1, \ldots$
            Append G to $Returns(S_t)$
            $V(S_t) \leftarrow \text{average}(Returns($

Monte Carlo ES (Exploring Starts), for estimating $\pi \approx \pi_*$

Initialize:
    $\pi(s) \in \mathcal{A}(s)$ (arbitrarily), for all $s \in \mathcal{S}$
    $Q(s,a) \in \mathbb{R}$ (arbitrarily), for all $s \in \mathcal{S}$, $a \in \mathcal{A}(s)$
    $Returns(s,a) \leftarrow$ empty list, for all $s \in \mathcal{S}$, $a \in \mathcal{A}(s)$

Loop forever (for each episode):
    Choose $S_0 \in \mathcal{S}$, $A_0 \in \mathcal{A}(S_0)$ randomly such that all pairs have probability > 0
    Generate an episode from S_0, A_0, following π: $S_0, A_0, R_1, \ldots, S_{T-1}, A_{T-1}, R_T$
    $G \leftarrow 0$
    Loop for each step of episode, $t = T-1, T-2, \ldots, 0$:
        $G \leftarrow \gamma G + R_{t+1}$
        Unless the pair S_t, A_t appears in $S_0, A_0, S_1, A_1 \ldots, S_{t-1}, A_{t-1}$:
            Append G to $Returns(S_t, A_t)$
            $Q(S_t, A_t) \leftarrow \text{average}(Returns(S_t, A_t))$
            $\pi(S_t) \leftarrow \arg\max_a Q(S_t, a)$

Learn policy

Q-Values instead of V.

Exploring starts

First-visit

Greedy policy update

Code...

MC Control without Exploring Starts

- **Issue**: The environment needs to let us start episodes at different (s, a) at will. This is fine for simulations, but hard in the real world.

- **Need**: For convergence, we need a guarantee that all (reachable) state-action combinations are tried.

- Options:
  a) **On-policy control**: learn a **soft policy** instead of the optimal policy.
  b) **Off-policy control**: use a soft **behavior policy** for sampling that tries all state-action pairs, but learn the optimal, deterministic **target policy**.

On-Policy Monte Carlo Control

Find a good policy while using it to sample.

Exploration with On-Policy Methods

On-policy methods learn the policy they follow.

Dilemma:

- We need behavior (follow a policy) that explores. Such a policy is called a soft policy.
- We know that MDPs have optimal deterministic policies that do not explore.

Optimality issue: On-policy methods can only learn soft policies. We can convert a soft policy into a greedy deterministic policy, but it may not be optimal!

Types of Soft Policies for Exploration

Definition : A policy is soft if
$$\pi(a|s) > 0 \qquad \forall s \in \mathcal{S}, a \in \mathcal{A}$$

Soft policies are stochastic and used as behavior policies that guarantee exploration.

Important Soft Policy Types:

- **ϵ-soft policy**: agent takes with probability ϵ a random action.

$$\pi(a|s) \geq \frac{\epsilon}{|\mathcal{A}(s)|}$$

- **ϵ-greedy policy**: follows a greedy policy but chooses a random action with probability ϵ

$$\pi(a|s) = \begin{cases} 1 - \epsilon + \dfrac{\epsilon}{\mathcal{A}(s)} & \text{if } a = \text{greedy action} \\ \dfrac{\epsilon}{|\mathcal{A}(s)|} & \text{if } a \neq \text{greedy action} \end{cases}$$

Alg. On-Policy MC Control with ϵ-soft Policies

Monte Carlo ES (Exploring Starts), for estimating $\pi \approx \pi_*$

Initialize:
  $\pi(s) \in \mathcal{A}(s)$ (arbitrarily), for all $s \in \mathcal{S}$
  $Q(s, a) \in \mathbb{R}$ (arbitrarily), for all $s \in \mathcal{S}$, $a \in \mathcal{A}$
  $Returns(s, a) \leftarrow$ empty list, for all $s \in \mathcal{S}$, a

Loop forever (for each episode):
  Choose $S_0 \in \mathcal{S}$, $A_0 \in \mathcal{A}(S_0)$ randomly such
  Generate an episode from S_0, A_0, fol
  $G \leftarrow 0$
  Loop for each step of episode, $t = T-1, T-$
    $G \leftarrow \gamma G + R_{t+1}$
    Unless the pair S_t, A_t appears in $S_0, A_0,$
      Append G to $Returns(S_t, A_t)$
      $Q(S_t, A_t) \leftarrow$ average$(Returns(S_t, A_t)$
      $\pi(S_t) \leftarrow \arg\max_a Q(S_t, a)$

On-policy first-visit MC control (for ε-soft policies), estimates $\pi \approx \pi_*$

Algorithm parameter: small $\varepsilon > 0$
Initialize:
  $\pi \leftarrow$ an arbitrary ε-soft policy
  $Q(s, a) \in \mathbb{R}$ (arbitrarily), for all $s \in \mathcal{S}$, $a \in \mathcal{A}(s)$
  $Returns(s, a) \leftarrow$ empty list, for all $s \in \mathcal{S}$, $a \in \mathcal{A}(s)$

Repeat forever (for each episode):
  Generate an episode following π: $S_0, A_0, R_1, \ldots, S_{T-1}, A_{T-1}, R_T$
  $G \leftarrow 0$
  Loop for each step of episode, $t = T-1, T-2, \ldots, 0$:
    $G \leftarrow \gamma G + R_{t+1}$
    Unless the pair S_t, A_t appears in $S_0, A_0, S_1, A_1 \ldots, S_{t-1}, A_{t-1}$:
      Append G to $Returns(S_t, A_t)$
      $Q(S_t, A_t) \leftarrow$ average$(Returns(S_t, A_t))$
      $A^* \leftarrow \arg\max_a Q(S_t, a)$ (with ties broken arbitrarily)
      For all $a \in \mathcal{A}(S_t)$:

$$\pi(a|S_t) \leftarrow \begin{cases} 1 - \varepsilon + \varepsilon/|\mathcal{A}(S_t)| & \text{if } a = A^* \\ \varepsilon/|\mathcal{A}(S_t)| & \text{if } a \neq A^* \end{cases}$$

Start with soft policy

No exploring starts!

Create ϵ-greedy policy

Important: On-policy learns the best ϵ-greedy policy, but not the optimal deterministic policy!

Convergence:
Greedy in the Limit with Infinite Exploration (GLIE)

Issues:

1. On-policy MC control algorithms change the policy and **average over returns from different policies**. This could be a problem for convergence!
2. On-policy methods learn a **soft policy** which is crucial for exploration.

GLIE Condition

Many on-policy control methods converge to the optimal deterministic policy if:

1. **Infinite explorations**: Every state-action pair is visited infinitely often.

$$\lim_{t \to \infty} N_t(s, a) = \infty$$

2. **Greedy in the limit**: The policy becomes greedy with respect to Q over time.

$$\lim_{t \to \infty} \pi\big(\mathrm{argmax}_a(Q_t(s, a), s)\big) = 1$$

Practical GLIE Approximation:

- Infinite exploration: Use many episodes and an ϵ-greedy policy for exploration. This will “wash out” early “bad” observations and makes sure every policy is considered.
- Greedy in the limit: Use an ϵ-greedy policy where ϵ is reduced over time. E.g., $\epsilon_t = \frac{1}{t}$

Off-Policy Methods

Prediction and Control

Off-Policy Prediction

Estimate a value function for a policy different from the sampling policy.

On-Policy vs. Off-Policy Control

On-policy: On-policy methods learn the policy they follow. These need to be exploring (soft).

Off-policy: Use two policies
- **Target policy**: learn π, which will become an optimal deterministic policy.
- **Behavior policy** b: guarantees exploration with a soft policy.

- Called “off-policy” because the generated data follows a different policy than the learned target policy.

- Off-policy methods are typically:
  - - More complicated
  - - Converge slower
  - + Often more powerful
  - + Can be applied to data externally generated (e.g., a training data set).

Note: On-policy learning is a special case of off-policy learning with $b = \pi$

Off-Policy Prediction

- **Goal**: Estimate v_π or q_π given episodes generated with $b \neq \pi$.

- **Assumptions for Convergence**:
  - b is fixed.
  - **Coverage**: Every action taken under π will occasionally also be taken under b

$$\pi(a|s) > 0 \Rightarrow b(a|s) > 0$$

  - That is, b needs to be stochastic (soft) for states where it differs from π (typically b is ϵ-greedy)

- **Issue**: Returns and sample averages are valid for b but not for π! We need to correct them.

Importance Sampling

- **Importance sampling** is a method to estimate the expected values under a desired distribution, given samples from another distribution
    - Expectation under desired distribution: $\mathbb{E}_\pi(G_t|S_t = s)$
    - Observed expectation from sampled distribution: $\mathbb{E}_b(G_t|S_t = s)$
- **Idea**: correct returns observed under b to estimate returns for π

- Probability of a trajectory starting at S_t under π

Needs unknown transition probabilities!

$$\Pr\{A_t, S_{t+1}, A_{t+1}, \dots, A_{T-1}, S_T | S_t, A_{t:T-1} \sim \pi\} = \prod_{k=t}^{T-1} \pi(A_k|S_k) p(S_{k+1}|S_k, A_k)$$

- **Importance sampling ratio:** How much more likely are the remaining actions in the sequence chosen under π than under b:

$$\rho_{t:T-1} \stackrel{\text{def}}{=} \frac{\prod_{k=t}^{T-1} \pi(A_k|S_k) \cancel{p(S_{k+1}|S_k, A_k)}}{\prod_{k=t}^{T-1} b(A_k|S_k) \cancel{p(S_{k+1}|S_k, A_k)}} = \prod_{k=t}^{T-1} \frac{\pi(A_k|S_k)}{b(A_k|S_k)}$$

Luckily, the unknown transition probabilities cancel out!

- Now we get:

$$v_\pi(s) = \mathbb{E}_\pi(G_t|S_t = s) = \mathbb{E}_b[\rho_{t:T-1} G_t | S_t = s]$$

- **Note**: We only get $\rho_{t:T-1} > 0$ when all A_k chosen by b have a $\pi(A_k|S_k) > 0$!

This is especially a problem if π is a deterministic policy very different from b!

Ordinary vs. Weighted Importance Sampling

- **Ordinary importance sampling** averages the corrected sample returns

$$V(s) \stackrel{\text{def}}{=} \frac{\sum_{t \in \mathcal{T}(s)} \rho_{t:T(t)-1} G_t}{|\mathcal{T}(s)|} \longrightarrow \mathbb{E}_b[\rho_{t:T-1} G_t | S_t = s] = \mathbb{E}_\pi(G_t | S_t = t)$$

  - $\mathcal{T}(s)$ are the time steps that s was (first) visited.
  - Gives an unbiased estimate of $v_\pi(s)$ but the variance is extreme for low $|\mathcal{T}(s)|$.

Note: the $\frac{|\mathcal{T}(s)|}{|\mathcal{T}(s)|}$ is missing since it canceled out.

- **Weighted importance sampling**

$$V(s) \stackrel{\text{def}}{=} \frac{\sum_{t \in \mathcal{T}(s)} \rho_{t:T(t)-1} G_t}{\sum_{t \in \mathcal{T}(s)} \rho_{t:T(t)-1}} \longrightarrow \frac{\mathbb{E}_b[\rho_{t:T-1} G_t | S_t = s]}{\mathbb{E}_b\left[\rho_{t:T(t)-1} \middle| S_t = s\right]} = \frac{\mathbb{E}_\pi(G_t | S_t = t)}{1}$$

  - Set 0 if denominator is 0.
  - Initially biased to $v_b(s)$. The ρ terms cancel for a single observation and G_t is sampled using b.
  - Bias falls asymptotically with more visits
  - Preferred method with much lower errors (variance) for smaller number of episodes.

Alg. Off-Policy Prediction

Weighted importance sampling + incremental average update

On-policy first-visit MC control (for ε-soft policies), estimates $\pi \approx \pi_*$

Algorithm parameter: small $\varepsilon > 0$
Initialize:
  $\pi \leftarrow$ an arbitrary ε-soft policy
  $Q(s, a) \in \mathbb{R}$ (arbitrarily), for all s
  $Returns(s, a) \leftarrow$ empty list, for a

Repeat forever (for each epis…
  Generate an episode follo…
  $G \leftarrow 0$
  Loop for each step of episode, t =
    $G \leftarrow \gamma G + R_{t+1}$
    Unless the pair S_t, A_t appears
      Append G to $Returns(S_t,$
      $Q(S_t, A_t) \leftarrow$ average($Retu$…
      $A^* \leftarrow \operatorname{argmax}_a Q(S_t, a)$
      For all $a \in \mathcal{A}(S_t)$:
        $\pi(a|S_t) \leftarrow \begin{cases} 1 - \varepsilon + \dots \\ \varepsilon / |\mathcal{A}(S\dots \end{cases}$

Off-policy MC prediction (policy evaluation) for estimating $Q \approx q_\pi$

Input: an arbitrary target policy π
Initialize, for all $s \in \mathcal{S}$, $a \in \mathcal{A}(s)$:
  $Q(s, a) \in \mathbb{R}$ (arbitrarily)
  $C(s, a) \leftarrow 0$

cumulative sum of weights for (s, a) for incremental update of the average return.

Loop forever (for each episode):
  $b \leftarrow$ any policy with coverage of π

b can change every episode! Practical choice is $b = \epsilon\text{−greedy}(\pi)$

  Generate an episode following b: $S_0, A_0, R_1, \ldots, S_{T-1}, A_{T-1}, R_T$
  $G \leftarrow 0$
  $W \leftarrow 1$

Importance sampling ratio ρ

  Loop for each step of episode, $t = T-1, T-2, \ldots, 0$, while $W \neq 0$:
    $G \leftarrow \gamma G + R_{t+1}$
    $C(S_t, A_t) \leftarrow C(S_t, A_t) + W$
    $Q(S_t, A_t) \leftarrow Q(S_t, A_t) + \frac{W}{C(S_t, A_t)} [G - Q(S_t, A_t)]$
    $W \leftarrow W \frac{\pi(A_t|S_t)}{b(A_t|S_t)}$

Can only learn from the end of the episode where the action chosen from b can also be chosen by π!

Incremental update + Correct the update

Off-Policy Control

Find a good policy while following a different behavior policy.

Off-Policy MC Control

- **Remember**: on-policy control tries to estimate a policy while using it for sampling. This has the following implications:
  - The **behavior changes** with the learned policy.
  - The policy that is learned needs to explore and therefore **may not be able to learn the optimal deterministic policy**.

- Off-policy control uses two policies:

  - **Behavior policy $\boldsymbol{b}$**: Needs to have coverage. That is, it eventually takes every action taken under π. This typically means that we will use a **soft policy**.

  - **Target policy $\boldsymbol{\pi}$**: Typically, a **deterministic greedy policy** which can become optimal.

- **Convergence**: With a covering behavior policy.

Alg. Off-Policy MC Control

Off-policy MC prediction (policy evaluation) for estimating $Q \approx q_\pi$

Input: an arbitrary target policy π
Initialize, for all $s \in \mathcal{S}$, $a \in \mathcal{A}(s)$:
    $Q(s,a) \in \mathbb{R}$ (arbitrarily)
    $C(s,a) \leftarrow 0$

Loop forever (for each episod
    $b \leftarrow$ any policy with cove
    Generate an episode followin
    $G \leftarrow 0$
    $W \leftarrow 1$
    Loop for each step of episode, t
        $G \leftarrow \gamma G + R_{t+1}$
        $C(S_t, A_t) \leftarrow C(S_t, A_t) + W$
        $Q(S_t, A_t) \leftarrow Q(S_t, A_t) + \frac{W}{C(S}$
        $W \leftarrow W \frac{\pi(A_t|S_t)}{b(A_t|S_t)}$

Off-policy MC control, for estimating $\pi \approx \pi_*$

Initialize, for all $s \in \mathcal{S}$, $a \in \mathcal{A}(s)$:
    $Q(s,a) \in \mathbb{R}$ (arbitrarily)
    $C(s,a) \leftarrow 0$
    $\pi(s) \leftarrow \arg\max_a Q(s,a)$ (with ties broken consistently)

Add the greedy target policy

Loop forever (for each episode):
    $b \leftarrow$ any soft policy
    Generate an episode using b: $S_0, A_0, R_1, \ldots, S_{T-1}, A_{T-1}, R_T$
    $G \leftarrow 0$
    $W \leftarrow 1$
    Loop for each step of episode, $t = T-1, T-2, \ldots, 0$:
        $G \leftarrow \gamma G + R_{t+1}$
        $C(S_t, A_t) \leftarrow C(S_t, A_t) + W$
        $Q(S_t, A_t) \leftarrow Q(S_t, A_t) + \frac{W}{C(S_t,A_t)} [G - Q(S_t, A_t)]$
        $\pi(S_t) \leftarrow \arg\max_a Q(S_t, a)$ (with ties broken consistently)
        If $A_t \neq \pi(S_t)$ then exit inner Loop (proceed to next episode)
        $W \leftarrow W \frac{1}{b(A_t|S_t)}$

Practical choice is $b = \epsilon\text{−greedy}(\pi)$

Can only use the end of the episode that uses greedy actions for π
Issue: Not very sample efficient!

We use π to select a. So, the probability is always 1.

What you Should Know

MC works with sampled episodes:

- **Policy Evaluation**: Averages many sampled returns.
- **Control**: Interleaves pol. eval. and pol. improvement on an episode-by-episode basis.
- **On vs. off-policy control.**
- **Convergence**: converge with GLIE or covering behavior policies. Convergence is typically slow.

Advantages of MC methods vs. DP:

- **Model-free**: No need for an environment model. Sampling from a simulation is often easy.
- Can evaluate a small **state set** of interest. DP always has to sweep over all states.
- **Unbiased estimates:** Estimates are based on returns from complete sampled sequences. We will learn about other methods that introduce bias by bootstrapping (estimating a value from other estimates).
- Using complete sequences may be less harmed by violations of the Markov property/issues with observability.

Issues:

- **Needs to maintain sufficient exploration**. Options are:
  1. Exploring starts can learn the optimal policy.
  2. On-policy control with an ϵ-soft policy and GLIE.
  3. Off-policy control with a covering behavior policy. Note: only a part of the data can be used.
- MC methods are **not sample efficient**. I.e., sampled episodes are used only once and then discarded.
- **Not a truly online method**: Updates are not done after each action but are delayed until the end of each episode.